THE PICARD SCHEME OF A CURVE DECOMPOSES INTO SMOOTH PROPER PIECES

THE STACKS PROJECT AUTHORS

Editorial note. Statement span is lines 1412-1432 of sources/stacks-pic/pinned-original-source.tex; proof span is lines 1434-1543, the complete author proof (108 body lines).

Editorial note. Navigation stubs: 11 cited-but-not-included labels are supplied as non-load-bearing pointers with their pinned tags where the snapshot tag index has them.

Editorial note. extra_packages is empty on purpose: the retained preamble already loads xy with options and a second declaration clashes in the assembler wrapper.

Editorial note. Text is retained verbatim, including the author's own typos and placeholders.

Editorial note. Queue index 0 of queue_pic.json (pb=4839, stmt 1412-1432, proof 1434-1543).

Editorial note. stacks-pic wave-1 (pi01): fresh adjudication accept (2026-10-14); pool per standing queue hypercovering -> pic (SMC reassigned to Zhen per Chief 08:5x); pic.tex 27 labels/23 with proof judged 21 accept / 2 reject / 4 pb=0 env-dup rows excluded; spans cite-free and reference/slogan/history-free so no patchenv needed. Named omission: Author omits the precise formulation and proof of the isomorphism Z = Proj(Sym(Gamma(X,L)^*)) - the inverse construction (section gives effective Cartier divisor) is described in-span; named mechanical check.

Allowed standard prerequisites (editorial).

- Explicit prerequisite boundary. The cited inputs are prerequisites; the argument in the retained span is not replaced by any cited result.

Source reference. Original lemma 00R4: sources/stacks-pic/pinned-original-source.tex, source lines 1543–1543. Cited by the retained statement or proof but not part of the collected spans; supplied only as a navigation pointer. The argument does not depend on its content.

Source reference. Original lemma 001P: sources/stacks-pic/pinned-original-source.tex, source lines 1543–1543. Cited by the retained statement or proof but not part of the collected spans; supplied only as a navigation pointer. The argument does not depend on its content.

Source reference. Original lemma 047L: sources/stacks-pic/pinned-original-source.tex, source lines 1543–1543. Cited by the retained statement or proof but not part of the collected spans; supplied only as a navigation pointer. The argument does not depend on its content.

Source reference. Original lemma 0B9X: sources/stacks-pic/pinned-original-source.tex, source lines 1214–1219. Cited by the retained statement or proof but not part of the collected spans; supplied only as a navigation pointer. The argument does not depend on its content.

Date: stacks-project:pic.tex:local-snapshot-20261012; GNU FDL 1.2 or later.

Source reference. Original lemma 0AH6: sources/stacks-pic/pinned-original-source.tex, source lines 1543–1543. Cited by the retained statement or proof but not part of the collected spans; supplied only as a navigation pointer. The argument does not depend on its content.

Source reference. Original lemma 01V8: sources/stacks-pic/pinned-original-source.tex, source lines 1543–1543. Cited by the retained statement or proof but not part of the collected spans; supplied only as a navigation pointer. The argument does not depend on its content.

Source reference. Original lemma 0B9T: sources/stacks-pic/pinned-original-source.tex, source lines 1543–1543. Cited by the retained statement or proof but not part of the collected spans; supplied only as a navigation pointer. The argument does not depend on its content.

Source reference. Original lemma 0B9I: sources/stacks-pic/pinned-original-source.tex, source lines 849–867. Cited by the retained statement or proof but not part of the collected spans; supplied only as a navigation pointer. The argument does not depend on its content.

Source reference. Original lemma 0B9Z: sources/stacks-pic/pinned-original-source.tex, source lines 1377–1380. Cited by the retained statement or proof but not part of the collected spans; supplied only as a navigation pointer. The argument does not depend on its content.

Source reference. Original lemma 0B9J: sources/stacks-pic/pinned-original-source.tex, source lines 912–931. Cited by the retained statement or proof but not part of the collected spans; supplied only as a navigation pointer. The argument does not depend on its content.

Source reference. Original lemma 0B90: sources/stacks-pic/pinned-original-source.tex, source lines 1543–1543. Cited by the retained statement or proof but not part of the collected spans; supplied only as a navigation pointer. The argument does not depend on its content.

Lemma 0.1. *Let k be a separably closed field. Let X be a smooth projective curve of genus g over k.*

1. *$\underline{\mathrm{Pic}}_{X/k}$ is a disjoint union of g-dimensional smooth proper varieties $\underline{\mathrm{Pic}}^d_{X/k}$,*
2. *k-points of $\underline{\mathrm{Pic}}^d_{X/k}$ correspond to invertible $\mathcal{O}_X$-modules of degree d,*
3. *$\underline{\mathrm{Pic}}^0_{X/k}$ is an open and closed subgroup scheme,*
4. *for $d \geq 0$ there is a canonical morphism $\gamma_d : \underline{\mathrm{Hilb}}^d_{X/k} \to \underline{\mathrm{Pic}}^d_{X/k}$*
5. *the morphisms γ_d are surjective for $d \geq g$ and smooth for $d \geq 2g-1$,*
6. *the morphism $\underline{\mathrm{Hilb}}^g_{X/k} \to \underline{\mathrm{Pic}}^g_{X/k}$ is birational.*

Proof. Pick a k-rational point σ of X. Recall that $\mathrm{Pic}_{X/k}$ is isomorphic to the functor $\mathrm{Pic}_{X/k,\sigma}$. By Derived Categories of Schemes, Lemma 0B9T for every $d \in \mathbf{Z}$ there is an open subfunctor

$$\mathrm{Pic}^d_{X/k,\sigma} \subset \mathrm{Pic}_{X/k,\sigma}$$

whose value on a scheme T over k consists of those $\mathcal{L} \in \mathrm{Pic}_{X/k,\sigma}(T)$ such that $\chi(X_t, \mathcal{L}_t) = d + 1 - g$ and moreover we have

$$\mathrm{Pic}_{X/k,\sigma} = \coprod\nolimits_{d \in \mathbf{Z}} \mathrm{Pic}^d_{X/k,\sigma}$$

as fppf sheaves. It follows that the scheme $\underline{\mathrm{Pic}}_{X/k}$ (which exists by Proposition 0B9Z) has a corresponding decomposition

$$\underline{\mathrm{Pic}}_{X/k,\sigma} = \coprod\nolimits_{d \in \mathbf{Z}} \underline{\mathrm{Pic}}^d_{X/k,\sigma}$$

where the points of $\underline{\mathrm{Pic}}^d_{X/k,\sigma}$ correspond to isomorphism classes of invertible modules of degree d on X.

Fix $d \geq 0$. There is a morphism

$$\gamma_d : \underline{\mathrm{Hilb}}^d_{X/k} \longrightarrow \underline{\mathrm{Pic}}^d_{X/k}$$

coming from the invertible sheaf $\mathcal{O}(D_{univ})$ on $\underline{\mathrm{Hilb}}^d_{X/k} \times_k X$ (Remark 0B9J) by the Yoneda lemma (Categories, Lemma 001P). Our proof of the representability of the Picard functor of X/k in Proposition 0B9Z and Lemma 0B9X shows that γ_g induces an open immersion on a nonempty open of $\underline{\mathrm{Hilb}}^g_{X/k}$. Moreover, the proof shows that the translates of this open by k-rational points of the group scheme $\underline{\mathrm{Pic}}_{X/k}$ define an open covering. Since $\underline{\mathrm{Hilb}}^g_{X/K}$ is smooth of dimension g (Proposition 0B9I) over k, we conclude that the group scheme $\underline{\mathrm{Pic}}_{X/k}$ is smooth of dimension g over k.

By Groupoids, Lemma 047L we see that $\underline{\mathrm{Pic}}_{X/k}$ is separated. Hence, for every $d \geq 0$, the image of γ_d is a proper variety over k (Morphisms, Lemma 0AH6).

Let $d \geq g$. Then for any field extension K/k and any invertible $\mathcal{O}_{X_K}$-module $\mathcal{L}$ of degree d, we see that $\chi(X_K, \mathcal{L}) = d + 1 - g > 0$. Hence $\mathcal{L}$ has a nonzero section and we conclude that $\mathcal{L} = \mathcal{O}_{X_K}(D)$ for some divisor $D \subset X_K$ of degree d. It follows that γ_d is surjective.

Combining the facts mentioned above we see that $\underline{\mathrm{Pic}}^d_{X/k}$ is proper for $d \geq g$. This finishes the proof of (2) because now we see that $\underline{\mathrm{Pic}}^d_{X/k}$ is proper for $d \geq g$ but then all $\underline{\mathrm{Pic}}^d_{X/k}$ are proper by translation.

It remains to prove that γ_d is smooth for $d \geq 2g - 1$. Consider an invertible $\mathcal{O}_X$-module $\mathcal{L}$ of degree d. Then the fibre of the point corresponding to $\mathcal{L}$ is

$$Z = \{D \subset X \mid \mathcal{O}_X(D) \cong \mathcal{L}\} \subset \underline{\mathrm{Hilb}}^d_{X/k}$$

with its natural scheme structure. Since any isomorphism $\mathcal{O}_X(D) \to \mathcal{L}$ is well defined up to multiplying by a nonzero scalar, we see that the canonical section $1 \in \mathcal{O}_X(D)$ is mapped to a section $s \in \Gamma(X, \mathcal{L})$ well defined up to multiplication by a nonzero scalar. In this way we obtain a morphism

$$Z \longrightarrow \mathrm{Proj}(\mathrm{Sym}(\Gamma(X, \mathcal{L})^*))$$

(dual because of our conventions). This morphism is an isomorphism, because given an section of $\mathcal{L}$ we can take the associated effective Cartier divisor, in other words we can construct an inverse of the displayed morphism; we omit the precise formulation and proof. Since $\dim H^0(X, \mathcal{L}) = d + 1 - g$ for every $\mathcal{L}$ of degree $d \geq 2g - 1$ by Varieties, Lemma 0B90 we see that $\mathrm{Proj}(\mathrm{Sym}(\Gamma(X, \mathcal{L})^*)) \cong \mathbf{P}^{d-g}_k$. We conclude that $\dim(Z) = \dim(\mathbf{P}^{d-g}_k) = d - g$. We conclude that the fibres of the morphism γ_d all have dimension equal to the difference of the dimensions of $\underline{\mathrm{Hilb}}^d_{X/k}$ and $\underline{\mathrm{Pic}}^d_{X/k}$. It follows that γ_d is flat, see Algebra, Lemma 00R4. As moreover the fibres are smooth, we conclude that γ_d is smooth by Morphisms, Lemma 01V8. $\square$